Swiss Institute of Bioinformatics

INTRODUCTION TO SEQUENCING DATA ANALYSIS

File types

`.fasta` `.fastq` `.sam/bam` `.bed` `.gff` `.vcf`

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Frequently used file types



.fasta

sequences

Raw sequences: nucleotides or proteins



.fastq

reads

Sequencing reads with per-base quality



.sam/bam

alignments

Reads aligned to a reference



.bed

regions

Genomic intervals, 0-based



.gff

annotations

Genes, transcripts, exons & features



.vcf

variants

Differences from the reference

Broad overview of file types: https://genome-euro.ucsc.edu/FAQ/FAQformat.html

fasta

FASTA

- Plain sequence: *.fasta or *.fa
- Nucleotides or amino acids (proteins)

Useful command

```
$ grep -c "^>" sequence.fasta
2
```

Counts the header lines, i.e. the number of sequences

sequence.fasta

```
>sequence title1
ATCGTATCTATCGTATCT
GGTTTATCGTATCT
>sequence title2
ATGATGACGT
```

- `>` header line: name / description of the sequence
- sequence, can span several lines

fastq files

FASTQ

reads.fastq

```
@D00283R:66:CC611ANXX:4:2311:2596:2330 1:N:0:TCCGGAG
ACTCTACGCTCAATAAAGATTTCTGATACGGCTCCTGAAATGCAGAATGAGT
+
B/<<<B<FFFFFFFFFBBFFFBFFFFBFFFF/FFFFFFFF/BFFFFFBFFF
```

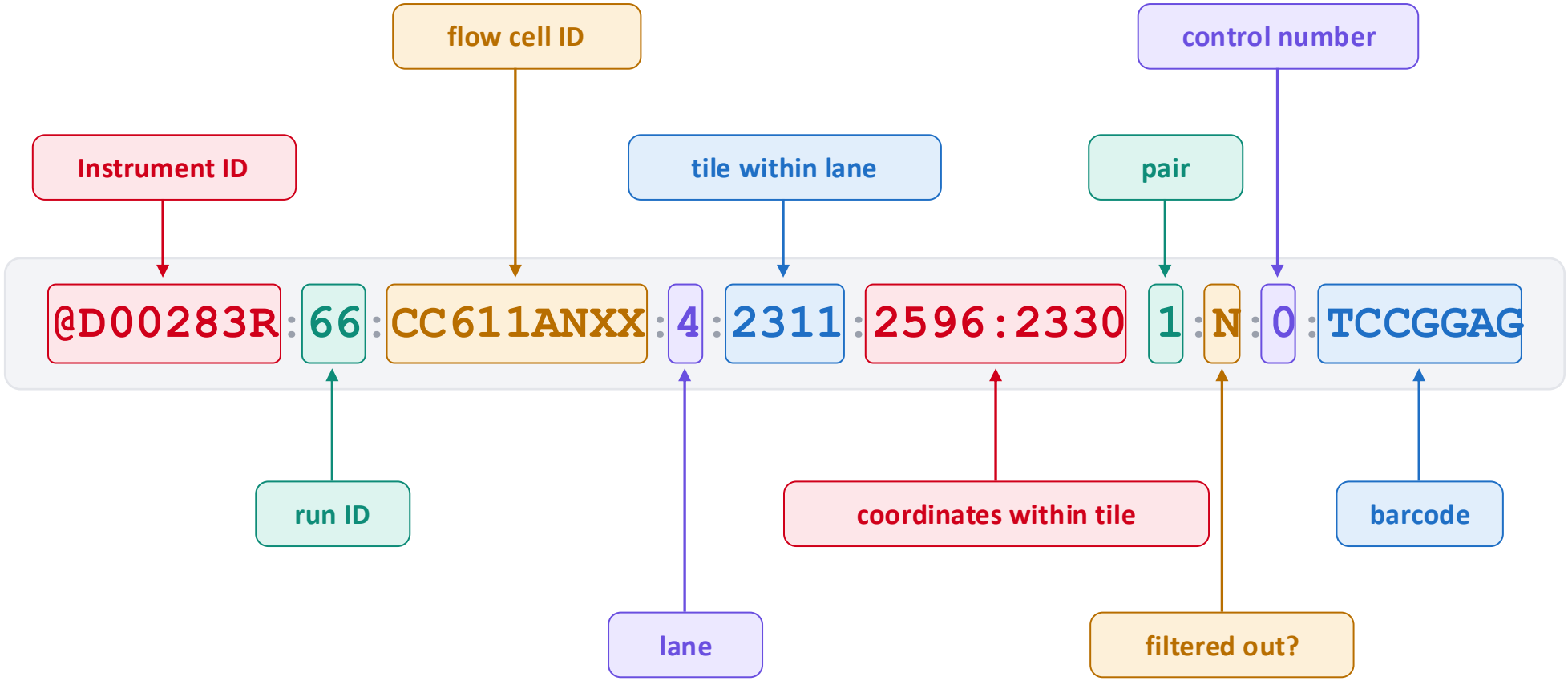
- title, starts with @
- nucleotide sequence
- optional description
- base quality

Base quality is encoded as one ASCII character per base (Phred+33)



fastq header

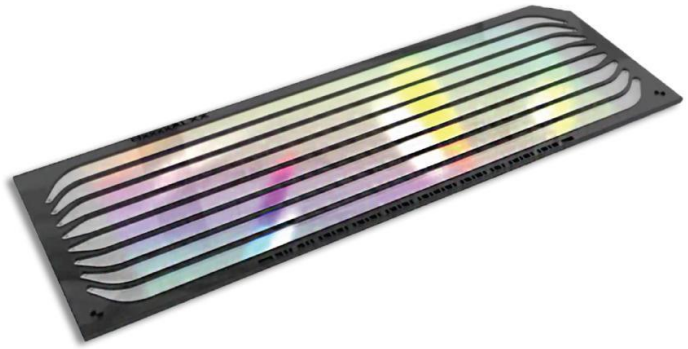
FASTQ



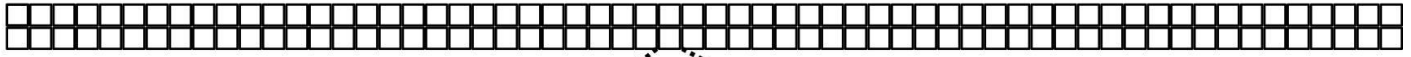
Structure of a flow cell

FASTQ

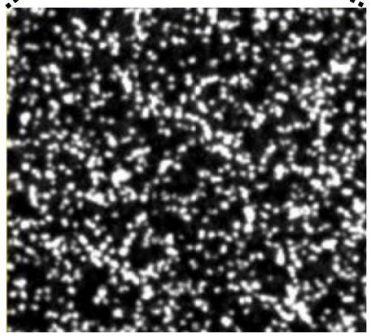
HiSeq flow cell
8 lanes



1 lane, 120 tiles



up to 5 billion
reads per flow cell
on Hiseq 4000



QUIZ

Question 9

Answer on Vevox



sam & bam: alignment files

SAM/BAM

.sam

sequence alignment format

- human-readable
- large files
- used for inspection / debugging



.bam

binary version of sam

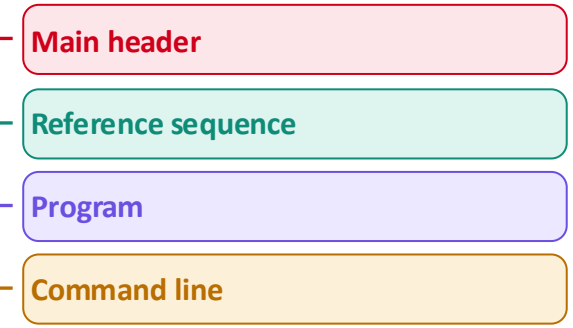
- not human-readable
- smaller & faster
- used for computation

sam header

SAM/BAM

alignments.sam

```
@HD  VN:1.5  SO:coordinate
@SQ  SN:U00096.3  LN:4641652
@PG  ID:bowtie2  PN:bowtie2  VN:2.4.1
CL:"/opt/miniconda3/envs/ngs/bin/bowtie2-align-s \
   --wrapper basic-0 \
   -x /home/ubuntu/ecoli/ref_genome//ecoli-strK12-MG1655.fasta \
   -1 /home/ubuntu/ecoli/trimmed_data/paired_trimmed_SRR519926_1.fastq \
   -2 /home/ubuntu/ecoli/trimmed_data/paired_trimmed_SRR519926_2.fastq"
```



SN sequence name **LN** sequence length

@SQ fields in human data

SAM/BAM

```
@HD     VN:1.4  SO:coordinate
@SQ     SN:chr1 LN:248956422
@SQ     SN:chr2 LN:242193529
@SQ     SN:chr3 LN:198295559
@SQ     SN:chr4 LN:190214555
@SQ     SN:chr5 LN:181538259
@SQ     SN:chr6 LN:170805979
@SQ     SN:chr7 LN:159345973
@SQ     SN:chr8 LN:145138636
@SQ     SN:chr9 LN:138394717
@SQ     SN:chr10        LN:133797422
@SQ     SN:chr11        LN:135086622
@SQ     SN:chr12        LN:133275309
@SQ     SN:chr13        LN:114364328
@SQ     SN:chr14        LN:107043718
@SQ     SN:chr15        LN:101991189
@SQ     SN:chr16        LN:90338345
@SQ     SN:chr17        LN:83257441
@SQ     SN:chr18        LN:80373285
@SQ     SN:chr19        LN:58617616
@SQ     SN:chr20        LN:64444167
@SQ     SN:chr21        LN:46709983
@SQ     SN:chr22        LN:50818468
@SQ     SN:chrX LN:156040895
@SQ     SN:chrY LN:57227415
@SQ     SN:chrM LN:16569
```

One @SQ line per reference sequence

25

reference sequences · chr1–22, chrX, chrY, chrM

248,956,422 bp

LN of chr1 · the longest human chromosome

`SN` sequence name `LN` sequence length

One aligned read from the E. coli dataset

SAM/BAM

alignments.sam · one line, tab-separated

```
@D00283R:66:CC611ANXX…   81   U00096.3   107   24   6M2I2M2I239M   =   116   261

GCTCTTCCGATCTTTAGGTCACTAAATACTTTAACCAATATAGGCATAGCGCACAGACAGATAAAAATTACAGAGTCCACAACATCCATGAAACGCATTAGCACCACCATTACCCCCACCATCACC
ATTACCACAGGTAACGGTGCGGGCTGCCGCGTACAGGAAACACAGAAAAAAGCCCGCACCTGACAGTGCGGGCTTTTTTTTTCGACCAAAGGTAACGAGGTAACAACCATGCGAGTGTTGAAGTT

###@<0>?4(A>:(4:>>:@>CB@>:>4@@>>8+@CCC:@4(>:3<0.7DCC>C>>::>>@8DCACC@::CCA::((+(A@4::4C>:>><<2@AACCA@><9?:+AC@>)&50&?:3CC>:>4C@
@::9?@:@CDBBB?BDDDDDDDDDBDDDDDCCCCDDDDCACCDDDDDDDDDDDDDCC>DDDDDDDDDDDDDEDDFHJJJJJIJJIJJJJJJJJJJIJIJJJJIHFJJJJJHHHHHFFFFFCCC

AS:i:-44  XN:i:0  XM:i:7  XO:i:2  XG:i:4  NM:i:11  MD:Z:0T0T1A4C63A37A37A98  YS:i:-57  YT:Z:DP
```

QNAME	FLAG	RNAME	POS	MAPQ	CIGAR	mate
read ID	read properties	reference	start position	alignment confidence	alignment structure	position & insert size

The same read, column by column

SAM/BAM

#	field	SAM column	example
1	QNAME	read name	@D00283R:66:CC61…
2	FLAG	flag	81
3	RNAME	reference	U00096.3
4	POS	start position	107
5	MAPQ	mapping quality	24
6	CIGAR	CIGAR string	6M2I2M2I239M
7	RNEXT	reference name mate is mapped	=
8	PNEXT	start position mate	116
9	TLEN	fragment length	261
10	SEQ	sequence	GCTCTTCCGA…
11	QUAL	base quality	###@<0>?4(…
12+	TAG	optional	AS:i:-44
	TAG	optional	XN:i:0

Optional tags explained

SAM/BAM

Tag	Meaning	Value	Notes
AS	Alignment score	-44	Lower = worse
XN	Ambiguous bases in ref (e.g. Ns)	0	None
XM	Mismatches	7	Aligner-specific
XO	Gap opens	2	2 insertions
XG	Gap bases total	4	2 insertions of 2
NM	Edit distance	11	7 mismatches + 4 indel bases
MD	Reference mismatches	0T0T1A4…	Reference-level
YS	Mate's alignment score	-57	Worse than this read
YT	Alignment type	DP	Discordant pair

CIGAR strings

SAM/BAM

Concise Idiosyncratic Gapped Alignment Report

Op	BAM	Description
M	0	alignment match (can be a sequence match or mismatch)
I	1	insertion to the reference
D	2	deletion from the reference
N	3	skipped region from the reference
S	4	soft clipping (clipped sequences present in SEQ)
H	5	hard clipping (clipped sequences NOT present in SEQ)
P	6	padding (silent deletion from padded reference)
=	7	sequence match
X	8	sequence mismatch

Almost never used

read vs reference

```
AAATGGG
|||  |||
AAACGGG
```

7M

classic: 7 aligned bases

3=1X3=

extended: 3 matches, 1 mismatch, 3 matches

https://www.biostars.org/p/17043/

https://www.biostars.org/p/354942/

https://samtools.github.io/hts-specs/SAMv1.pdf

QUIZ

Question 10

Answer on Vevox



sam flags

SAM/BAM

Bit		Description
1	0x1	template having multiple segments in sequencing
2	0x2	each segment properly aligned according to the aligner
4	0x4	segment unmapped
8	0x8	next segment in the template unmapped
16	0x10	SEQ being reverse complemented
32	0x20	SEQ of the next segment in the template being reverse complemented
64	0x40	the first segment in the template
128	0x80	the last segment in the template
256	0x100	secondary alignment
512	0x200	not passing filters, such as platform/vendor quality controls
1024	0x400	PCR or optical duplicate
2048	0x800	supplementary alignment

flags add up as bits

```
1   ->   001
2   ->   010
4   ->   100
_____________
3   ->   011   (1 + 2)
```

1	0	Read is paired
2	1	Read is mapped in proper pair
4	2	Read is unmapped
8	3	Mate is unmapped
16	4	Read is mapped to reverse strand
32	5	Mate is mapped to reverse strand
64	6	This is the first read in a pair
128	7	This is the second read in a pair

Segment ≈ read

https://samtools.github.io/hts-specs/SAMv1.pdf

Adding up flags: three examples

SAM/BAM

	read paired? +1	properly aligned? +2	unmapped? +4	mate unmapped? +8	flag
read1	1	1	0	0	3 1 + 2
read2	1	0	0	1	9 1 + 8
read3	1	0	1	1	13 1 + 4 + 8

flag = sum of the bit values of the properties that are true

QUIZ

Question 11

Answer on Vevox

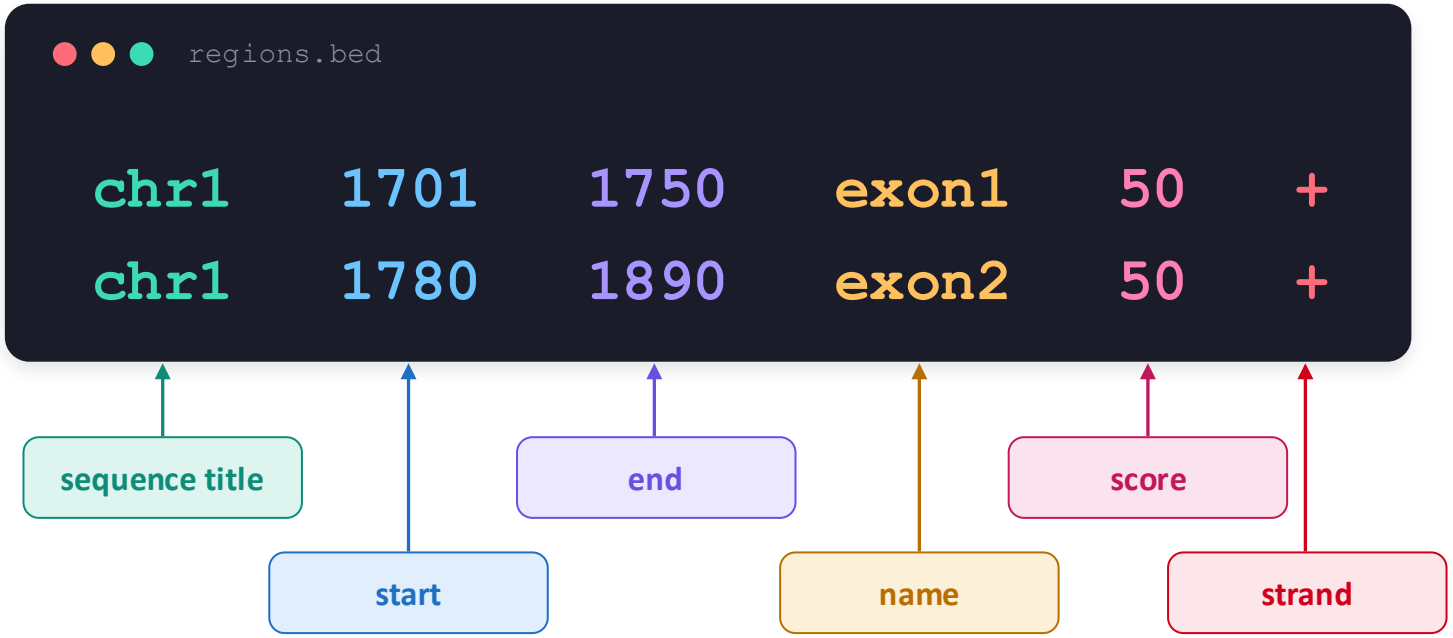


bed

BED

Browser **E**xtensible **D**ata

Aim: specify regions



Numbering starts at 0!

The same regions, written 1-based:

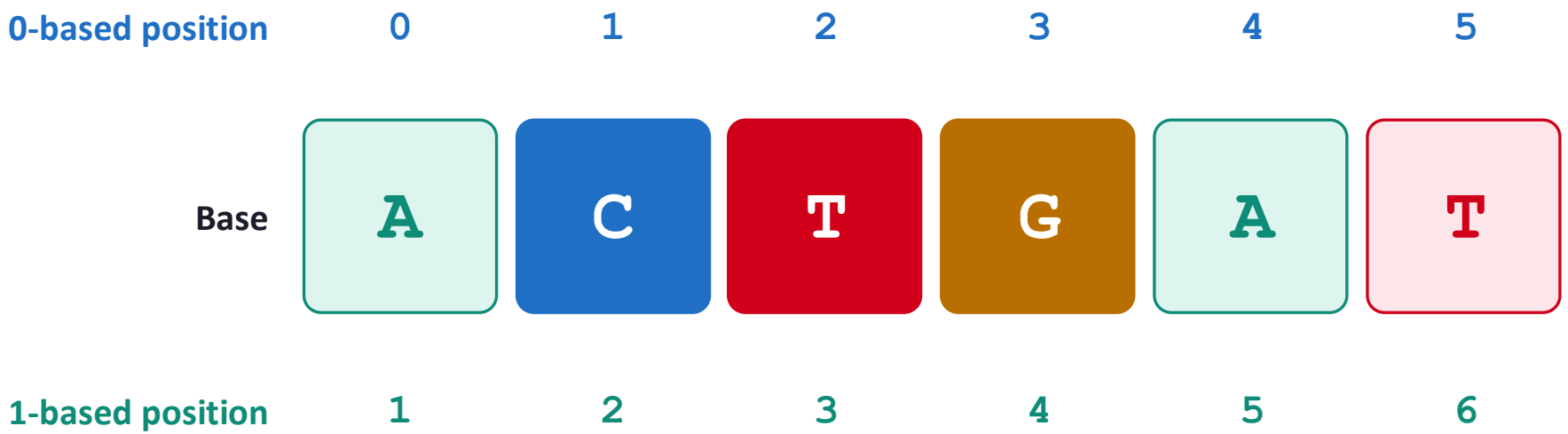
`chr1:1702-1750`

`chr1:1781-1890`

Simple DNA sequence example

COORDS

Sequence: **ACTGAT**



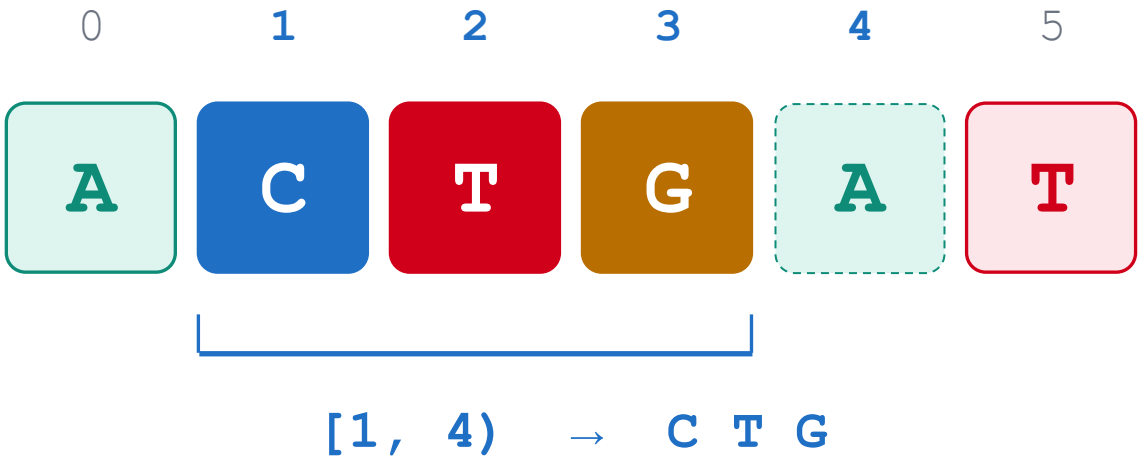
Same bases, two ways of counting — C T G are used in the next two examples

The 0-based system

COORDS

Used by: BED, BigBed, Python, C, etc.

- The first nucleotide (A) is at position 0
- Intervals are [start, end): start included, end excluded



Example

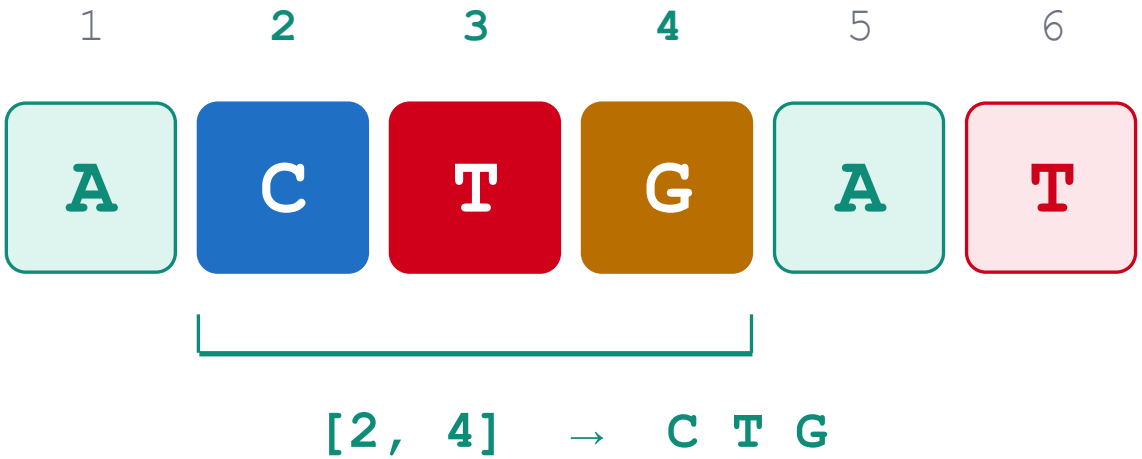
A BED interval with start = 1, end = 4 → positions 1, 2, 3 → C, T, G

The 1-based system

COORDS

Used by: GTF/GFF, VCF, SAM/BAM

- The first nucleotide (A) is at position 1
- Intervals are [start, end]: start and end included



Example

A GTF interval with start = 2, end = 4 → positions 2, 3, 4 → C, T, G

QUIZ

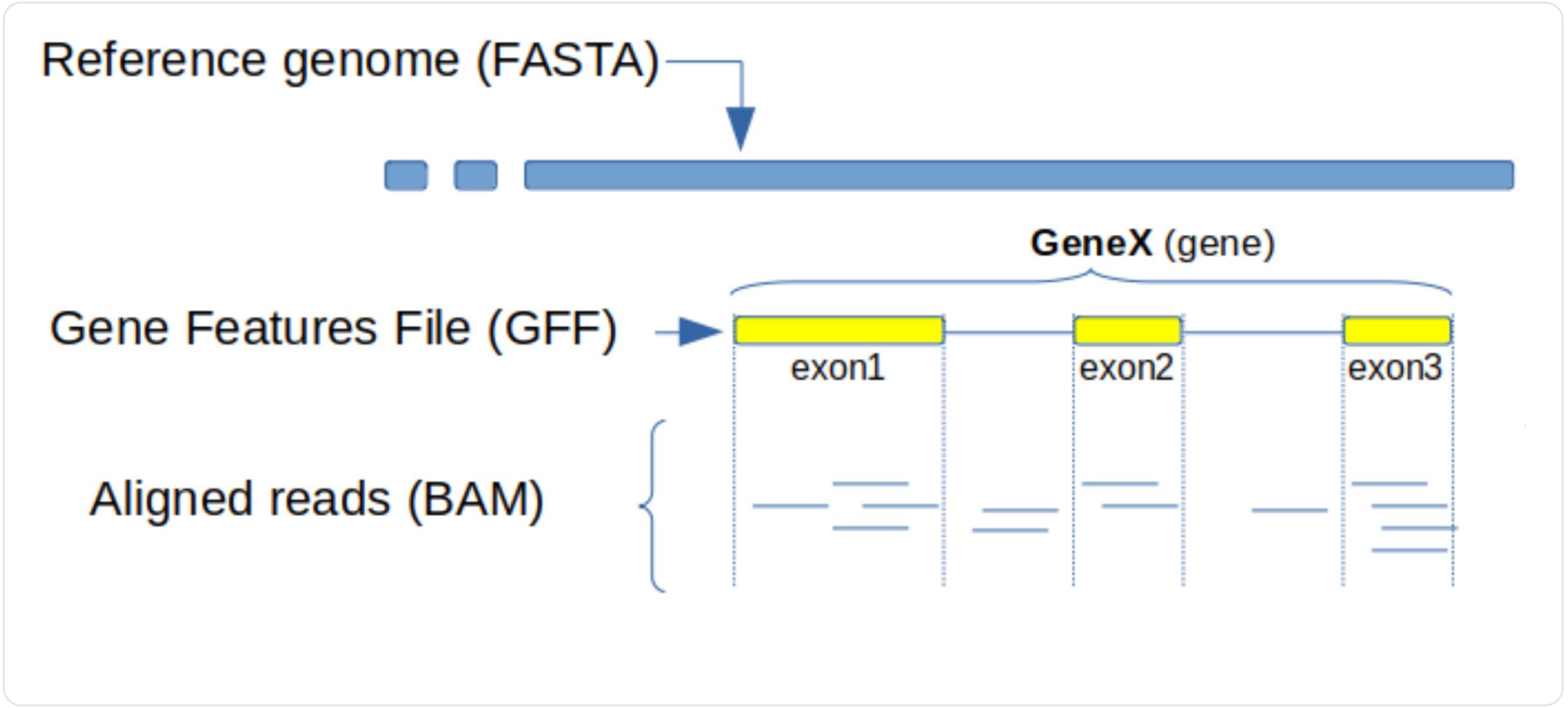
Question 12

Answer on Vevox



gff: where annotations fit

GFF



From https://bioinfogp.cnb.csic.es/tools/seqnjoy/help/

gff / gtf

GTF = flat and gene-centric. GFF3 = hierarchical and general.

GFF

General Feature Format

Aim: annotation · 9 tab-separated columns

1 seq name	2 source	3 feature	4 start	5 end	6 score	7 strand	8 frame	9 attributes
1	ensembl	mRNA	339070	346959	.	-	.	ID=…;
1	ensembl	exon	339070	339312	.	-	.	Parent=…;
1	ensembl	CDS	339070	339312	.	-	0	ID=…;

start & end are 1-based and inclusive · feature types: gene, mRNA / transcript, exon, CDS, start_codon …

A real GFF3 file: GENCODE v49

GFF3

gencode.v49.annotation.gff3 · ELOVL5 (chr6)

```
##gff-version 3
#description: evidence-based annotation of the human genome (GRCh38), version 49 (Ensembl 115)
#provider: GENCODE
#contact: gencode-help@ebi.ac.uk
#format: gff3
#date: 2025-07-08
…

chr6 HAVANA gene         53260285 53349179 . - . ID=ENSG00000012660.15;gene_id=ENSG00000012660.15;gene_type=protein_coding;…
chr6 HAVANA transcript   53260285 53348973 . - . ID=ENST00000954545.1;Parent=ENSG00000012660.15;gene_id=ENSG00000012660.15;…
chr6 HAVANA exon         53348817 53348973 . - . ID=exon:ENST00000954545.1:1;Parent=ENST00000954545.1;gene_id=ENSG000000126…
chr6 HAVANA exon         53295642 53295707 . - . ID=exon:ENST00000954545.1:2;Parent=ENST00000954545.1;gene_id=ENSG000000126…
chr6 HAVANA CDS          53295642 53295699 . - 0 ID=CDS:ENST00000954545.1;Parent=ENST00000954545.1;gene_id=ENSG00000012660.…
chr6 HAVANA start_codon  53295697 53295699 . - 0 ID=start_codon:ENST00000954545.1;Parent=ENST00000954545.1;gene_id=ENSG0000…
chr6 HAVANA exon         53291776 53291963 . - . ID=exon:ENST00000954545.1:3;Parent=ENST00000954545.1;gene_id=ENSG000000126…
```

Column 9 of the gene line: key=value attributes, separated by ;

ID=ENSG00000012660.15	gene_id=ENSG00000012660.15	gene_type=protein_coding	gene_name=ELOVL5
level=2	hgnc_id=HGNC:21308	tag=overlapping_locus	havana_gene=OTTHUMG00000016249.2

vcf

VCF

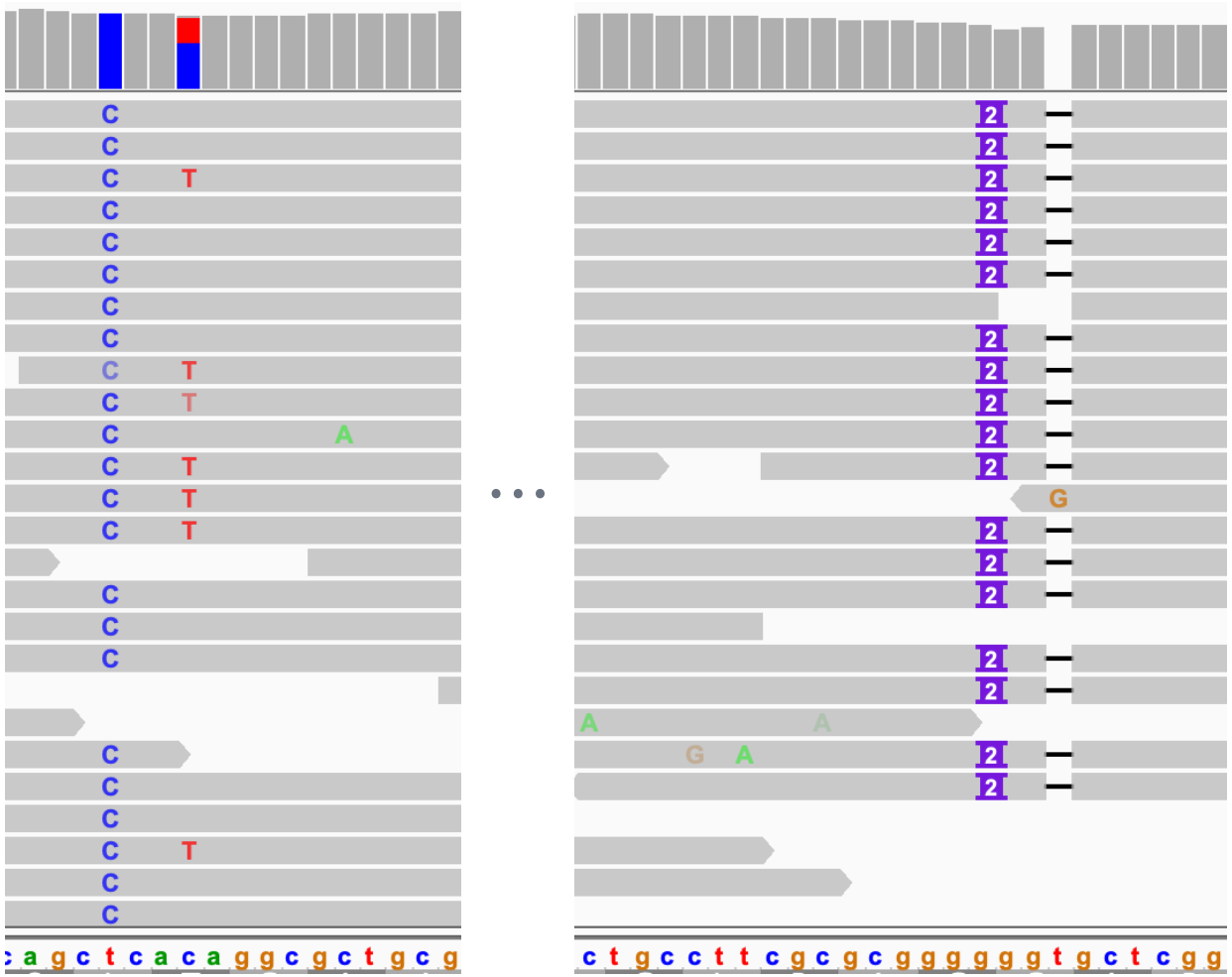
Variant Call Format

Aim: variants

```
##fileformat=VCFv4.2
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS     ID        REF    ALT     QUAL FILTER INFO                              FORMAT      NA00001        NA00002        NA00003
20     14370   rs6054257 G      A       29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51 1/1:43:5:.,.
20     17330   .         T      A       3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3   0/0:41:3
20     1110696 rs6040355 A      G,T     67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2   2/2:35:4
20     1230237 .         T      .       47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51 0/0:61:2
20     1234567 microsat1 GTC    G,GTCT  50   PASS   NS=3;DP=9;AA=G                    GT:GQ:DP    0/1:35:4       0/2:17:2       1/1:40:3
```

Variants seen in aligned reads

VCF



Genome browser view (IGV)

1 **Grey bars**
reads aligned to the reference

2 **Coloured letters**
bases that differ from the reference

3 **Same mismatch in many reads**
a candidate variant, reported in the vcf

vcf: header, columns and samples

VCF

```
##fileformat=VCFv4.3
##fileDate=20090805
##source=myImputationProgramV3.1
##reference=file:///seq/references/1000GenomesPilot-NCBI36.fasta
##contig=<ID=20,length=62435964,assembly=B36,md5=f126cdf8a6e0c7f379d618ff66beb2da,species="Homo sapiens",taxonomy=x>
##phasing=partial
##INFO=<ID=NS,Number=1,Type=Integer,Description="Number of Samples With Data">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Total Depth">
##INFO=<ID=AF,Number=A,Type=Float,Description="Allele Frequency">
##INFO=<ID=AA,Number=1,Type=String,Description="Ancestral Allele">
##INFO=<ID=DB,Number=0,Type=Flag,Description="dbSNP membership, build 129">
##INFO=<ID=H2,Number=0,Type=Flag,Description="HapMap2 membership">
##FILTER=<ID=q10,Description="Quality below 10">
##FILTER=<ID=s50,Description="Less than 50% of samples have data">
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype">
##FORMAT=<ID=GQ,Number=1,Type=Integer,Description="Genotype Quality">
##FORMAT=<ID=DP,Number=1,Type=Integer,Description="Read Depth">
##FORMAT=<ID=HQ,Number=2,Type=Integer,Description="Haplotype Quality">
#CHROM POS      ID         REF  ALT    QUAL FILTER INFO                              FORMAT      NA00001        NA00002
20     14370    rs6054257  G    A      29   PASS   NS=3;DP=14;AF=0.5;DB;H2           GT:GQ:DP:HQ 0|0:48:1:51,51 1|0:48:8:51,51
20     17330    .          T    A      3    q10    NS=3;DP=11;AF=0.017               GT:GQ:DP:HQ 0|0:49:3:58,50 0|1:3:5:65,3
20     1110696  rs6040355  A    G,T    67   PASS   NS=2;DP=10;AF=0.333,0.667;AA=T;DB GT:GQ:DP:HQ 1|2:21:6:23,27 2|1:2:0:18,2
20     1230237  .          T    .      47   PASS   NS=3;DP=13;AA=T                   GT:GQ:DP:HQ 0|0:54:7:56,60 0|0:48:4:51,51
20     1234567  microsat1  GTC  G,GTCT 50   PASS   NS=3;DP=9;AA=G                   GT:GQ:DP    0/1:35:4       0/2:17:2
```

INFO variant-level annotations

FILTER PASS or failed filter

FORMAT per-sample fields

samples one column per sample